SCHOOL OF COMPUTING AND DATA SCIENCES
FLAME UNIVERSITY
PUNE - 412115, MAHARASHTRA, INDIA

MoR for Interpretable Recursive Allocation in Language, Translation, and Medical Vision

A Graduation Project

Submitted by

DON DAS

For the award of the degree

Of

B.SC. HONOURS IN COMPUTER SCIENCE

April 2026

SCHOOL OF COMPUTING AND DATA SCIENCES
FLAME UNIVERSITY
PUNE - 412115, MAHARASHTRA, INDIA

MoR for Interpretable Recursive Allocation in Language, Translation, and Medical Vision

A Graduation Project

Submitted by

DON DAS

For the award of the degree

Of

B.SC. HONOURS IN COMPUTER SCIENCE

April 2026

© 2026 FLAME University

THESIS CERTIFICATE

This is to undertake that the Graduation Project titled **MOR FOR INTERPRETABLE RECURSIVE ALLOCATION IN LANGUAGE, TRANSLATION, AND MEDICAL VISION** , submitted by me to the School of Computing and Data Sciences, FLAME University, for the award of **B.SC. HONOURS IN COMPUTER SCIENCE**, is a bona fide record of the research work done by me under the supervision of **Prof. Sandeep Khanna**. The contents of this Graduation Project, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Pune - 412115, Maharashtra, India **Don Das**

Date: April 2026

Prof. Sandeep Khanna
Research advisor
Professor
School of Computing and Data Sciences
FLAME University

Prof. Jayarman V.K
Research co-advisor
Professor
School of Computing and Data Sciences
FLAME University

© 2026 FLAME University

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my research advisor, Prof. Sandeep Khanna, for his continuous guidance, encouragement, and insightful discussions throughout the course of this graduation project. His perspective on research methodology and academic writing has played a crucial role in shaping both the direction and quality of this work.

I am also deeply thankful to Prof. Jayarman V. K. for his valuable suggestions and academic support during different stages of the project. His inputs helped me refine several technical and conceptual aspects of the study.

I extend my appreciation to the faculty members of the School of Computing and Data Sciences, FLAME University, for providing a stimulating academic environment and the necessary institutional support for carrying out this research.

ABSTRACT

KEYWORDS Mixture of Recursions; adaptive computation; recursive depth; token routing; language models; interpretability; encoder-decoder; machine translation; medical image classification; skin lesion classification; HAM10000; IWSLT17; BLEU; entropy; transformer; deep learning; natural language processing; computer vision

This thesis presents a research study on Mixture of Recursions (MoR) with two connected goals. The first goal is explanatory; it aims to build and evaluate an analysis module that tests what exactly the recursive depth represents in a trained MoR system. The second goal focuses on extending MoR from its main language-model setting to two more tasks: encoder-decoder machine translation task and a medical image classification.

The explanatory module focuses on token-level routing analysis for a 3-recursion language model run. It processes 102,400 token observations and combines basic statistics, non-parametric tests, counterfactual probes, and consistency checks. The analysis shows a mixed result: while several intuitive MoR claims are supported, the important claim that deeper recursion directly tracks higher pre-recursion uncertainty is not supported in this run. Specifically, Spearman(depth, pre-recursion entropy) equals -0.1389, and Spearman(depth, entropy reduction) equals -0.2466. Deeper recursions are linked to lower-frequency tokens, and routing appears almost position-independent.

The machine translation extension introduces a hybrid encoder-decoder variant with a vanilla encoder and MoR decoder, BLEU-based checkpoint management, and early stopping hooks. Translation outcomes are functional but unstable across settings. This includes a method for monitoring check pointing using BLEU; there are early stop hooks for the experiment as well.

The medical vision paradigm extends MoR to HAM10000 skin-lesion classification with

a sparse expert-style routing. In patch-level analysis I observed non-trivial depth-diversity (mean depth of 1.750 across sampled diagnostics) and checkpoint metadata records a best validation accuracy up to 0.7596 for the sparse-from-scratch run.

The contribution of this thesis is very clear: best to consider MoR as an adaptive-computation family whose utility rests upon three interleaved tiers; (1) routing mechanism design, (2) optimization stability and (3) evidence discipline. The explanatory module which I developed here represents a reusable methodology to separate supported interpretability claims from attractive but unsupported narratives.

CONTENTS

		Page
ABSTRACT		**ii**
LIST OF TABLES		**vii**
LIST OF FIGURES		**viii**
GLOSSARY		**x**
ABBREVIATIONS		**xi**
NOTATION		**xii**
CHAPTER 1	**INTRODUCTION**	**1**
1.1	Motivation	2
1.2	Problem Statement	3
1.3	Research Questions	3
1.4	Thesis Structure	3
CHAPTER 2	**RELATED WORK**	**5**
2.1	Positioning This Study Within Adaptive Computation Research	6
CHAPTER 3	**PROPOSED FRAMEWORK**	**7**
3.1	Conceptual Model	7
3.2	Interpretability Hypothesis Space	8
3.3	Cross-Domain Transfer Logic	8
3.4	Expected Outcomes and Decision Logic	8
CHAPTER 4	**RESEARCH CONTRIBUTIONS**	**9**
4.1	Contribution Significance	9
4.2	Practical Contribution to the Project	10
CHAPTER 5	**METHODOLOGY**	**11**
5.1	Evidence Sources	11
5.2	Analysis Strategy	11
5.3	Validity Orientation	11
5.4	Statistical and Interpretive Safeguards	12
5.5	Counterfactual Logic as Mechanism Test	12
5.6	Reliability Across Task Types	12

CHAPTER 6 MODEL ARCHITECTURE 13

6.1 Language-Model Routing Study Context 13
6.2 Machine Translation Architecture 13
6.3 Medical Vision Architecture 16
6.4 Architectural Rationale and Trade-Offs 20
6.5 Capacity and Sharing Considerations 21

CHAPTER 7 LOSS FUNCTION 22

7.1 Core Task Losses 22
7.1.1 Language Modeling Branch (Autoregressive) 22
7.1.2 Machine Translation Branch (Seq2Seq with Label Smoothing) 23
7.1.3 Vision Branch (Classification) 23
7.2 Routing-Related Auxiliary Terms 23
7.2.1 MT Extension: Decoder Gate Regularization 23
7.2.2 LM MoR Auxiliary Terms (Project-Used Variants) 24
7.2.3 Vision MoR Auxiliary Terms 26
7.3 Gate Constraints and Stabilizers 26
7.4 Why Loss Design Is Central to Explanation 26

CHAPTER 8 EXPERIMENTAL SETUP 28

8.1 Language Modeling Explanatory Experiment 28
8.2 Machine Translation Experiments 28
8.3 Medical Vision Experiments 28
8.4 Reproducibility and Traceability Practices 29

CHAPTER 9 DATASETS 30

9.1 Language Data Context 30
9.2 Machine Translation Databases 30
9.3 Medical Vision Database 30
9.4 Database Characteristics and Their Interpretive Impact 30

CHAPTER 10 EXPERIMENTAL SETTINGS 32

10.1 System and Hardware Configuration 32
10.2 Language-Model Routing Settings 32
10.3 Translation Settings 32
10.4 Vision Settings 33
10.5 Setting Choices and Their Consequences 33

CHAPTER 11 RESULTS AND ANALYSIS 35

11.1 Language-Model Explanatory Results 35
11.1.1 Depth Allocation Structure 35
11.1.2 Correlation Outcomes 35
11.1.3 Category and Context Signals 37
11.1.4 Router Separation Strength 38

11.1.5 Counterfactual and Consistency 39
11.2 Machine Translation Results 40
11.2.1 Machine Translation Results and Interpretation 40
11.3 Quantitative Medical Vision Results 41
11.3.1 Baseline and Pretraining Comparisons 41
11.3.2 Recursion Depth Ablation Study 42
11.3.3 Vision Explainability Interpretation 43
11.4 Integrated Cross-Domain Reading of the Results 47
11.5 Why the Vision Signal Is Especially Important 48
11.6 Practical Interpretation of Translation Variance 49
11.7 Strength of the Language Explanatory Module 49

CHAPTER 12 DISCUSSION 50

12.1 Main Interpretation 50
12.2 Claim-Level Summary 50
12.3 Why the Vision Result Is Strong for Explanation 50
12.4 Limitations 52
12.5 Practical Next Steps 52

CHAPTER 13 CONCLUSION 53

BIBLIOGRAPHY 55

LIST OF TABLES

Table	Caption	Page
9.1	Summary of datasets used across the three experimental domains.	31
11.1	Depth allocation counts under MoR capacity constraints.	35
11.2	Correlation of recursion depth with uncertainty and token-level factors.	35
11.3	Mean entropy profile by assigned depth.	36
11.4	Mean depth by token category.	37
11.5	Selected vs. unselected router-score separation at deeper routing steps.	38
11.6	Frequency-only counterfactual agreement with actual routing.	39
11.7	Counterfactual confusion matrix (rows: actual depth, columns: frequency-only depth).	39
11.8	Per-depth agreement with frequency-only heuristic.	39
11.9	Within-token depth consistency (modal-depth fraction).	39

LIST OF FIGURES

Figure	Caption	Page
6.1	Schematic overview of the hybrid encoder–decoder architecture illustrating the vanilla Transformer encoder, the Mixture-of-Recursions (MoR) based decoder, and the cross-attention connections between them. The encoder processes the source sequence using a standard stacked self-attention structure, while the decoder refines target representations through recursively applied shared blocks with adaptive depth. Cross-attention layers enable information flow from encoder outputs to the decoder at each stage of recursive computation.	15
6.2	Router behaviour illustrating adaptive recursive computation in the MoR decoder, where simpler tokens are skipped at deeper recursion steps while more complex tokens remain fully engaged in successive decoding refinements.	16
6.3	End-to-end vision architecture of the Mixture-of-Recursions (MoR) model, illustrating the pipeline from ViT-style image patch embedding through a recursively shared MoR transformer backbone with expert-choice routing, to global pooled feature aggregation and final seven-class HAM10000 classification output	19
6.4	Adaptive router behaviour in the vision MoR backbone where informative image patch tokens receive deeper recursive processing while less salient patches are skipped.	20
11.1	Relationship between assigned recursion depth and token-level entropy measures. The plot compares pre-recursion entropy, final entropy, and entropy reduction across depth groups. Contrary to the common assumption that higher uncertainty directly leads to deeper recursion, the observed pattern shows that depth is not monotonically aligned with pre-recursion entropy. Instead, deeper recursion is associated with tokens that remain difficult even after refinement, indicating that routing decisions reflect a combination of lexical priors and contextual signals rather than a single uncertainty metric.	37
11.2	Average recursion depth across different token categories, including alphabetic, numeric, punctuation, whitespace, and special tokens. The results show that linguistically meaningful tokens (e.g., alphabetic tokens) receive deeper recursive processing on average, while structurally predictable elements such as whitespace and special tokens are processed at shallower depths. This supports the hypothesis that routing captures token-level complexity beyond simple frequency effects, aligning partially with predictability-based interpretations of adaptive computation.	38

11.3 Comparison of actual routing decisions with predictions from simple heuristic models, including entropy-only, frequency-only, and logistic combinations of features. The figure shows that while frequency-based heuristics provide some predictive power, they fail to fully explain routing behavior. The gap between heuristic performance and actual routing indicates that the model leverages additional contextual and learned signals, supporting the claim that MoR routing is non-trivial and not reducible to simple token-level statistics. 40
11.4 Patch-level recursion depth map for Sample 1434. The visualization illustrates the model's adaptive compute allocation: the concentrated, distinct core of the lesion receives lower recursion depth (dark blue), indicating high model confidence and visual homogeneity. In contrast, the model allocates higher recursive steps (green and yellow) to the surrounding boundaries and transition zones, signaling localized uncertainty and a demand for feature refinement where the disease merges with healthy tissue. 43
11.5 Patch-level recursion depth map for Sample 521. This sample further demonstrates the refinement-demand behavior of the MoR routing mechanism. Computational effort is minimized within the primary disease concentration due to its predictable structure, while the ambiguous texture transitions and peripheral areas around the lesion trigger deeper recursion to resolve structural uncertainty. 44
11.6 Original image 46
11.7 Ablation of maximum recursion depth on patch-level routing dynamics. As the allowed recursion budget increases from 2 to 5, the model resolves finer spatial structures, achieving sharper core isolation (dark blue) and concentrating the highest recursion depths on the most ambiguous boundary and transition textures. 47

GLOSSARY

Adaptive Computation	A modelling paradigm in which computational effort is dynamically allocated based on input complexity or uncertainty.
Auxiliary Routing Loss	Additional optimisation terms introduced to stabilise or regularise routing behaviour in adaptive neural networks.
Cross-Domain Transfer	The process of adapting a modelling framework from one task domain, such as language modelling, to other domains like machine translation or medical vision.
Mixture of Recursions (MoR)	An adaptive neural network framework that allocates variable recursive computation depth to tokens or patches using routing mechanisms.
Model Interpretability	The extent to which internal model behaviour and decision signals can be understood or meaningfully analysed by humans.
Patch-Level Depth Mapping	Spatial allocation of recursive computation depth across image patches in vision transformer-style architectures.
Recursive Depth	The number of recursive refinement steps applied to a token, feature vector, or image patch during model inference.
Routing Mechanism	A learned decision process that determines whether additional recursive processing should be applied to a representation.
Token-Level Depth Allocation	Assignment of different computational depths to individual sequence tokens based on routing decisions.

ABBREVIATIONS

AI Artificial Intelligence.

BLEU Bilingual Evaluation Understudy.

CUDA Compute Unified Device Architecture.

GPU Graphics Processing Unit.

HAM10000 Human Against Machine dataset (skin lesion dataset name).

IWSLT International Workshop on Spoken Language Translation.

LM Language Model.

MLP Multi-Layer Perceptron.

MoR Mixture of Recursions.

NLP Natural Language Processing.

ViT Vision Transformer.

NOTATION

β	Percentile threshold controlling token continuation in recursive routing
Φ	Parameters of the shared recursive transformation block
θ	Learnable parameters of the routing mechanism
d	Hidden feature dimension of model representations
$f(\cdot)$	Shared recursive transformation function in the MoR architecture
g_t^r	Routing gate value controlling recursive update at step r
H_t^r	Hidden representation of token or patch t at recursion step r
L	Overall training loss function
L_{aux}	Auxiliary routing regularisation loss
L_{task}	Primary task loss (e.g., language modelling, translation, or classification)
N_{tok}	Number of valid tokens used in loss computation
$p_\theta(\cdot)$	Model probability distribution parameterised by θ
R	Maximum number of recursive computation steps

CHAPTER 1

INTRODUCTION

A lot of modern deep learning systems consider the relationship between input and compute is fixed. For example, each token, patch, or example is given the same level of processing. This assumption is good for engineering, but it doesn't fit with how real data is structured. In language modeling, predictable continuations and infrequent ambiguous continuations coexist within the same sequence, a phenomenon highlighted by the scaling behavior of Transformer-based architectures Vaswani *et al.* (2017). In vision tasks, there are both uninformative background areas and small discriminative areas that determine class identity. This is what makes patch-based transformer processing strategies essential Dosovitskiy *et al.* (2021). It is therefore not good to give the same level of computing power to inputs of different levels of difficulty, and this may make it harder to see signals that help with interpretation.

A solution to this issue is proposed as Mixture of Recursions (MoR), which introduces adaptive recursive depth controlled through routing mechanisms Bae *et al.* (2025). MoR doesn't have a fixed forward depth. Instead, it lets transformations happen over and over again using the same set of transformer layers, and the routing logic decides how much recursive refinement each representational unit gets. Related concepts of adaptive computation and sparse routing have been explored in Mixture-of-Experts-based large language model scaling methodologies Zhang *et al.* (2024). The design has two parts: dynamic compute allocation and allocation signals that are interpretable. If depth assignment is linked to input uncertainty or semantic difficulty, then depth itself can be used to understand how the model works. This thesis looks into this question using an explanatory evaluation framework instead of just a confirmatory empirical narrative.

The dissertation further extends the exisitng MoR framework to two more application domains. The first extension focuses on machine translation, implementing and testing the compatibility of MoR with encoder–decoder architecture and introduce sequence-level evaluation metrics like BLEU. The second extension looks at medical visual recognition using the HAM10000 dermatoscopic skin lesion dataset Tschandl *et al.* (2018).

The contribution of this thesisis methodological and empirical:

1. A concrete explanatory module for routing behavior.
2. A claim-level audit of interpretability hypotheses.
3. Two task extensions that stress MoR under different data geometries.
4. A disciplined synthesis of what can and cannot be concluded from current evidence.

1.1 MOTIVATION

Adaptive computation has become increasingly important for three reasons.

1. Compute budgets are finite while model sizes continue to grow.
2. The difficulty of tokens varies.
3. For practical usage, both efficiency and reliable behavior examples are required.

MoR deals with all three . The actual question is whether routing choices correspond to useful structure and whether those choices are useful outside the training setting it was developed.

The effective computational load per training step can be lower than that of a fixed-depth transformer because recursive blocks reuse parameters and routing cuts down on the number of tokens that need full-depth processing. This could reduce training time and memory usage.

1.2 PROBLEM STATEMENT

The core problem addressed in this thesis is:

> Is it possible to interpret recursive depth assignment in MoR as a form of meaningful adaptive allocation, and can this be applied also in machine translation and medical imaging without resulting in unstable or uncertain behavior?

1.3 RESEARCH QUESTIONS

The study is organized around the following research questions.

RQ1. What does recursion depth correlate with in token-level language-model behavior?

RQ2. Which interpretability claims are fully supported, partially supported, or contradicted in the actual runs?

RQ3. Can MoR be integrated into encoder-decoder translation models while preserving trainability and evaluation traceability?

RQ4. Can MoR be adapted to medical image classification with measurable routing diversity and competitive validation behavior?

RQ5. What methodological standards are necessary to make adaptive routing claims scientifically reliable?

1.4 THESIS STRUCTURE

The thesis begins with conceptual and methodological grounding, then presents the explanatory module and its findings, then covers Machine Translation and vision

extensions, and finally synthesizes lessons, limitations, and a forward program.

CHAPTER 2

RELATED WORK

MoR is part of a larger group of adaptive computation that try to use more or less computation depending upon how complex the input is. Early exit and confidence-triggered inference methods dynamically decrease depth for simpler inputs, as seen in adaptive inference frameworks like Adaptive Computation Time Graves (2016) and multi-exit deep networks Teerapittayanon *et al.* (2016). Token pruning and sparse attention or routing methods distribute selective computation across spatial or sequential positions, particularly in transformer architectures optimized for efficient scaling Vaswani *et al.* (2017); Dosovitskiy *et al.* (2021).

Scaling based on Mixture-of-Experts additionally contributes conditional parameter activation, which routes each input to specialized expert subnetworks to make better utilization of resources and increase efficiency Shazeer *et al.* (2017); Zhang *et al.* (2024). Recursive or weight-sharing transformer architectures, cover dynamic recursive depth formulations like Mixture-of-Recursions, balance adaptable representational refinement with parameter efficiency Bae *et al.* (2025).

The original MoR formulation prioritized the dynamic depth allocation through routing as a solution to practical adaptive inference bottlenecks while maintaining predictive quality Bae *et al.* (2025). This thesis presents a novel contribution to the field of adaptive modeling literature by providing an explanatory evaluation approach and a systematic analysis of cross-task transferability, which separates it from previous works.

Most of the current adaptive computation literature evaluates models mainly through endpoint accuracy or throughput benchmarks. This thesis emphasizes on claim-level interpretability validation by analyzing the alignment of routing behavior with proposed

uncertainty signals, semantic complexity, and contextual dependencies. This distinction is important as adaptive architectures can demonstrate robust aggregate performance but the computation distribution might be difficult to interpret semantically.

2.1 POSITIONING THIS STUDY WITHIN ADAPTIVE COMPUTATION RESEARCH

Some adaptive compute models decide whether to keep depth for an entire sample, while others choose at the token or patch level. Models like Mixture of Experts send the input to different expert , but keep the depth the same. MoR is at an intriguing middle ground: it keeps a recursive path open but adjusts the effect of recursive steps through routing logic. This gives the model a built-in way to understand itself because it is always making decisions about how much depth to give each tokens.

The routing signal can be sharp without having any meaning. It can be balanced without being aware of how difficult it is. It can correlate with frequency but overlook contextual difficulty. This thesis prioritizes methodology over narrative. The architecture is inspired by related works, but this study contributes it by systematizing the testing of the resultant behavior.

CHAPTER 3

PROPOSED FRAMEWORK

The suggested framework in this dissertation is made up of two interconnected layers. The explanatory layer creates a quantitative module that uses token-level records, statistical tests, counterfactual probes, and consistency analysis to check how MoR routes language modeling. The extension layer moves MoR to two new tasks: medical vision and machine translation. It then looks at both tasks' performance and how interpretable the compute distribution is.

The framework does not assume interpretability. It tests interpretability.

3.1 CONCEPTUAL MODEL

At a high level, recursive update behavior follows:

$$\mathcal{H}_t^{r+1} = \begin{cases} g_t^r \, f(\mathcal{H}_t^r, \Phi') + \mathcal{H}_t^r, & \text{if } g_t^r > P_\beta(G^r) \\ \mathcal{H}_t^r, & \text{otherwise} \end{cases} \tag{3.1}$$

$$g_t^r = \mathcal{G}\left(\theta_r^\top \mathcal{H}_t^r\right) \tag{3.2}$$

where $\mathcal{H}_t^r$ denotes the hidden state of token t at recursion step r, $f(\cdot)$ represents the shared recursive transformation parameterized by Φ', and g_t^r is a scalar routing score produced by a lightweight router with parameters θ_r and non-linear activation function $\mathcal{G}(\cdot)$ (e.g., sigmoid or tanh). $P_\beta(G^r)$ denotes the β-percentile threshold over routing scores at recursion step r, which determines whether a token continues to receive further recursive computation or exits early.

3.2 INTERPRETABILITY HYPOTHESIS SPACE

Instead of testing a single broad claim, the explanatory module tests several smaller ones. This looks into whether deeper recursion means more uncertainty, whether deeper recursion means lower-frequency or otherwise harder lexical conditions, whether routing is context-sensitive beyond token identity, and whether router separation is clear and useful when there are capacity limits.

3.3 CROSS-DOMAIN TRANSFER LOGIC

Machine translation tests MoR when it comes to conditional sequence generation, where cross-attention and adequacy constraints make routing more difficult. Medical vision tests MoR amidst spatial uncertainty and class imbalance, where explanatory depth maps could be clinically significant.

3.4 EXPECTED OUTCOMES AND DECISION LOGIC

The framework doesn't assume that all hypotheses will pass. Instead, it expects a mixed outcome, with some claims getting strong support, some getting weak support, and some being contradicted. This is not a flaw. It is what a high-resolution audit is supposed to do. The framework's value is in turning interpretability narratives into clear categories of evidence that can be used to improve design.

CHAPTER 4

RESEARCH CONTRIBUTIONS

This dissertation makes several concrete contributions. It reframes MoR work as explanatory adaptive-computation research rather than tool-centered reporting, introduces a routing analysis module for language modeling with distributional, correlational, inferential, counterfactual, and context-dependence diagnostics, and builds a claim-level evidence matrix that separates supported, partially supported, and unsupported hypotheses. It also delivers two operational extensions, encoder-decoder machine translation and HAM10000 medical vision classification, followed by an evidence-driven discussion of what MoR currently explains well and where stability, metadata consistency, and calibration still need improvement.

4.1 CONTRIBUTION SIGNIFICANCE

The contribution significance is best understood in terms of research maturity. Early-stage adaptive systems often show either promising behavior with weak explanation or rich explanation with limited transfer. This work contributes to both fronts at once. It provides a concrete explanatory method with claim-level outputs, while also demonstrating that MoR ideas can be operationalized in two additional tasks beyond the original language-model emphasis.

The second important contribution is epistemic discipline. By explicitly preserving unsupported hypotheses, the dissertation avoids confirmation bias and provides a clearer foundation for future work. This is particularly important for adaptive computation where persuasive visual narratives can otherwise overshadow contradictory quantitative evidence.

4.2 PRACTICAL CONTRIBUTION TO THE PROJECT

From a practical standpoint, the thesis consolidates a project-wide interpretation layer. It links routing diagnostics, translation checkpoint logic, and vision behavior into one coherent argument. This creates immediate utility for subsequent experiments because future runs can be evaluated against the same explanatory criteria rather than ad hoc impressions.

CHAPTER 5

METHODOLOGY

The methodology is empirical and artifact-grounded, using recorded outputs rather than hypothetical claims.

5.1 EVIDENCE SOURCES

Three evidence streams are used: token-level routing analysis outputs for language modeling, translation training and evaluation logs with BLEU checkpoints, and vision diagnostic notebooks with checkpoint metadata.

5.2 ANALYSIS STRATEGY

The study applies a multi llevel audit pipeline that first begins with sanity checks and descriptive distributions, progresses through correlation analysis utilizing Spearman and Pearson statistics, conducts non-parametric group tests using Mann-Whitney comparisons and effect sizes, and concludes with token-category and context-feature analysis, router score separation analysis, frequency-only counterfactual assignment, and within-token consistency.

5.3 VALIDITY ORIENTATION

Methodological validity is treated as central. Unsupported hypotheses are explicitly labeled, statistical significance is interpreted with effect-size caution, and missing metadata is reported as a limitation rather than ignored.

5.4 STATISTICAL AND INTERPRETIVE SAFEGUARDS

This method relies heavily on two important safety measures. The first is the difference between being able to mathematically detect something and its practical meaning. When sample sizes are large, even very small effects can lead to p-values that are very small. The study thus analyzes rank and effect sizes along with significance values, thus avoiding the exaggeration of claims based solely on significance. The second protection is consistency across multiple views. A claim is considered more robust when it is present in multiple analytical perspectives, such as both correlation direction and categorical level behavior.

5.5 COUNTERFACTUAL LOGIC AS MECHANISM TEST

The frequency only counterfactual component works as a test for mechanism isolation. If frequency were the primary factor for determining the routing depth, then a simple assignment based on frequency would be very close to the actual depth. The low agreement and low kappa show that routing isn't just sorting by frequency, even though frequency does play a role. This is a helpful middle ground: simple priors are important, but they don't fully explain behavior.

5.6 RELIABILITY ACROSS TASK TYPES

Different tasks require distinct approaches to methodological reliability. In language analysis, high volume token records and repeated token identity statistics make the results more reliable. In translation, reliability is mainly influenced by checkpoint policy and evaluation scheduling, as BLEU can vary with decoding settings. In vision, reliability relies on split reproducibility and metadata completeness, as interpretability claims require both quantitative bases and coherent visual diagnostics.

CHAPTER 6

MODEL ARCHITECTURE

6.1 LANGUAGE-MODEL ROUTING STUDY CONTEXT

The main explanatory analysis for the language model takes place in the original language-model setting of Mixture of Recursions (MoR). For this study, the base MoR architecture is used exactly as it is, with no changes to the core mechanism's structure. The setup uses a token-choice router that can handle three recursion steps while still being limited by capacity. By looking at 102,400 token-level routing traces taken from FineWeb-Edu data, the evaluation shows how the model's native routing works, with capacity-balanced depth behavior still clearly visible in the resulting allocation distributions.

6.2 MACHINE TRANSLATION ARCHITECTURE

The MT extension uses a hybrid encoder-decoder MoR variant.This means that it uses a vanilla encoder with full unique layers, a MoR decoder with recursive refinement, cross-attention-aware routing, and a middle-cycle sharing structure for decoder recursion.

The architectural settings are a hidden dimension of 576, 29 encoder layers, and an effective decoder depth of 29 layers, which is obtained by recursive unrolling. The recursion count is set to 3, and *middle-cycle parameter sharing* is employed along the decoder path.

In the middle-cycle sharing strategy, the first and last layers of the decoder have their own parameters, but the middle layers make a shared recursion block that reuses its parameters over and over again. Concretely, if the effective depth is obtained by unrolling N_r recursive applications of a shared block, the model preserves full-capacity transformations at the input and output boundaries while tying weights only within the internal layers. This

design balances parameter efficiency and representational capacity, allowing deeper effective computation without proportionally increasing the number of unique trainable parameters.

At a high level, recursive update behaviour follows:

$$h_{r+1} = h_r + g_r \odot \big(f_r(h_r, c) - h_r\big), \qquad g_r \in [0, 1].$$

Here, h_r is the representation at recursion step r, f_r is the recursive transformation, c is context such as encoder states in translation, and g_r is a learned gate controlling recursive contribution.

The recursive gated decoder design outlined here has a conceptual resemblance to the expert-choice routing mechanism employed in Mixture-of-Recursions models, which is likewise the routing strategy utilized in the medical image classification task. In both cases, a learned gating signal dynamically controls whether a representation should undergo additional refinement steps, effectively allocating more ‘ assigns each token to a fixed recursion depth at the beginning through a single routing decision, determining its full computational path upfront; this avoids iterative selection but can introduce load-balancing challenges because different depths may receive uneven numbers of tokens.

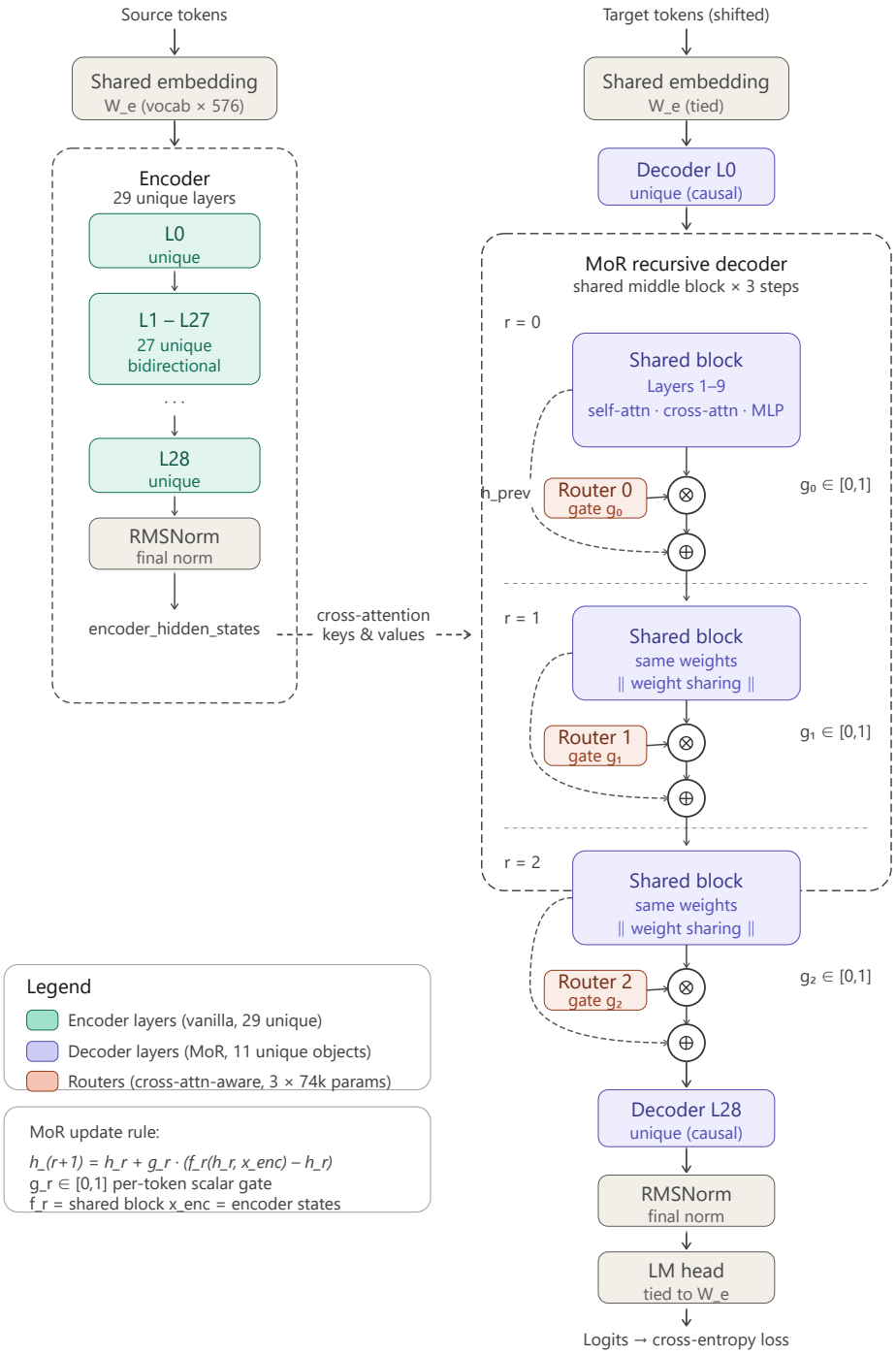


Figure 6.1: Schematic overview of the hybrid encoder–decoder architecture illustrating the vanilla Transformer encoder, the Mixture-of-Recursions (MoR) based decoder, and the cross-attention connections between them. The encoder processes the source sequence using a standard stacked self-attention structure, while the decoder refines target representations through recursively applied shared blocks with adaptive depth. Cross-attention layers enable information flow from encoder outputs to the decoder at each stage of recursive computation.

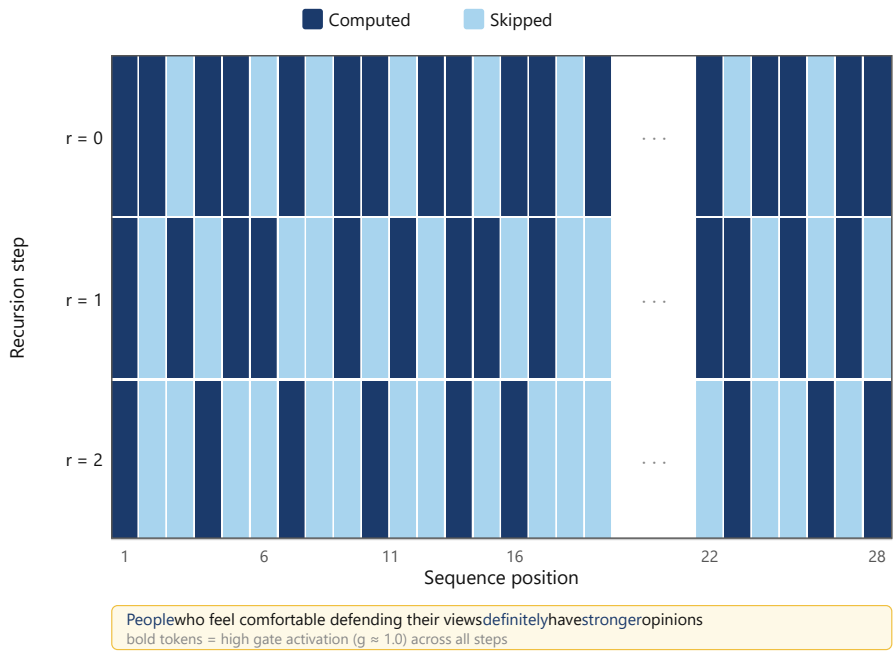


Figure 6.2: Router behaviour illustrating adaptive recursive computation in the MoR decoder, where simpler tokens are skipped at deeper recursion steps while more complex tokens remain fully engaged in successive decoding refinements.

6.3 MEDICAL VISION ARCHITECTURE

The vision extension keeps the original MoR transformer backbone and only changes the input interface, masking behavior, and output head to work for dermatology classification. The language modeling backbone is kept as the main computation block. Then, it is wrapped for vision so that image patch tokens can be sent through `inputs_embeds`, causal masking is removed, and a classifier takes the place of the LM decoding head.

Input formation is ViT-style patch embedding. For an RGB image $x \in \mathbb{R}^{B\times 3\times H\times W}$, a Conv2D projection with kernel size and stride equal to patch size P converts non-overlapping patches into hidden vectors of dimension d:

$$z = \mathrm{Conv2D}_{k=P,\, s=P}(x) \in \mathbb{R}^{B\times d\times H/P\times W/P}, \qquad t = \mathrm{flatten}(z) \in \mathbb{R}^{B\times N\times d}, \qquad N = \frac{H}{P}\cdot\frac{W}{P}. \tag{6.1}$$

A learnable positional embedding is then added to each patch token, yielding the final patch sequence passed to MoR. In the main HAM10000 configuration ($H = W = 224$, $P = 16$), this yields $N = 196$ patch tokens per image.

The recursive backbone behavior is kept from MoR, but attention is made bidirectional for vision. This means that attention modules are set to non-causal mode and a zero 4D attention mask is given so that all patches can pay attention to all others. The main difference between this and autoregressive text mode is that it keeps the recursive routing logic the same but gets rid of left-to-right constraints that don't make sense for images. Routing follows an **expert-choice mechanism**, where each recursion depth selects the most informative patch tokens to continue deeper computation. Formally, at recursion step r a routing score is computed for token (patch) t as

$$g_{r,t} = \sigma\left(\theta_r^\top h_{r,t}\right), \tag{6.2}$$

and only the top-k_r tokens are processed by the shared recursive block. The recursive update is therefore

$$h_{r+1,t} = \begin{cases} g_{r,t}\, f\left(h_{r,t};\Phi'\right) + h_{r,t}, & \text{if } g_{r,t} > \tau_r, \\ h_{r,t}, & \text{otherwise,} \end{cases} \tag{6.3}$$

where τ_r is the capacity-dependent routing threshold enforcing expert-choice selection.

At the output side, the language-model head is replaced by a classification head. Final hidden states are globally averaged across patch positions, normalized (RMSNorm), regularized by dropout, and projected to seven logits corresponding to HAM10000 classes (`akiec`, `bcc`, `bkl`, `df`, `mel`, `nv`, `vasc`). Training uses cross-entropy classification

loss together with optional MoR auxiliary routing losses, ensuring that routing remains active while optimizing diagnosis-level prediction.

Major vision runs also use sparse expert-style routing schedules. A representative funnel capacity profile $(1.0,\ 0.5,\ 0.25)$ means all 196 patches are processed at recursion 1, the top 98 continue to recursion 2, and the top 49 continue to recursion 3 (with warmup in sparse settings). This converts routing into a spatially interpretable compute-allocation signal, enabling patch-depth explanation analyses rather than only image-level confidence reporting.

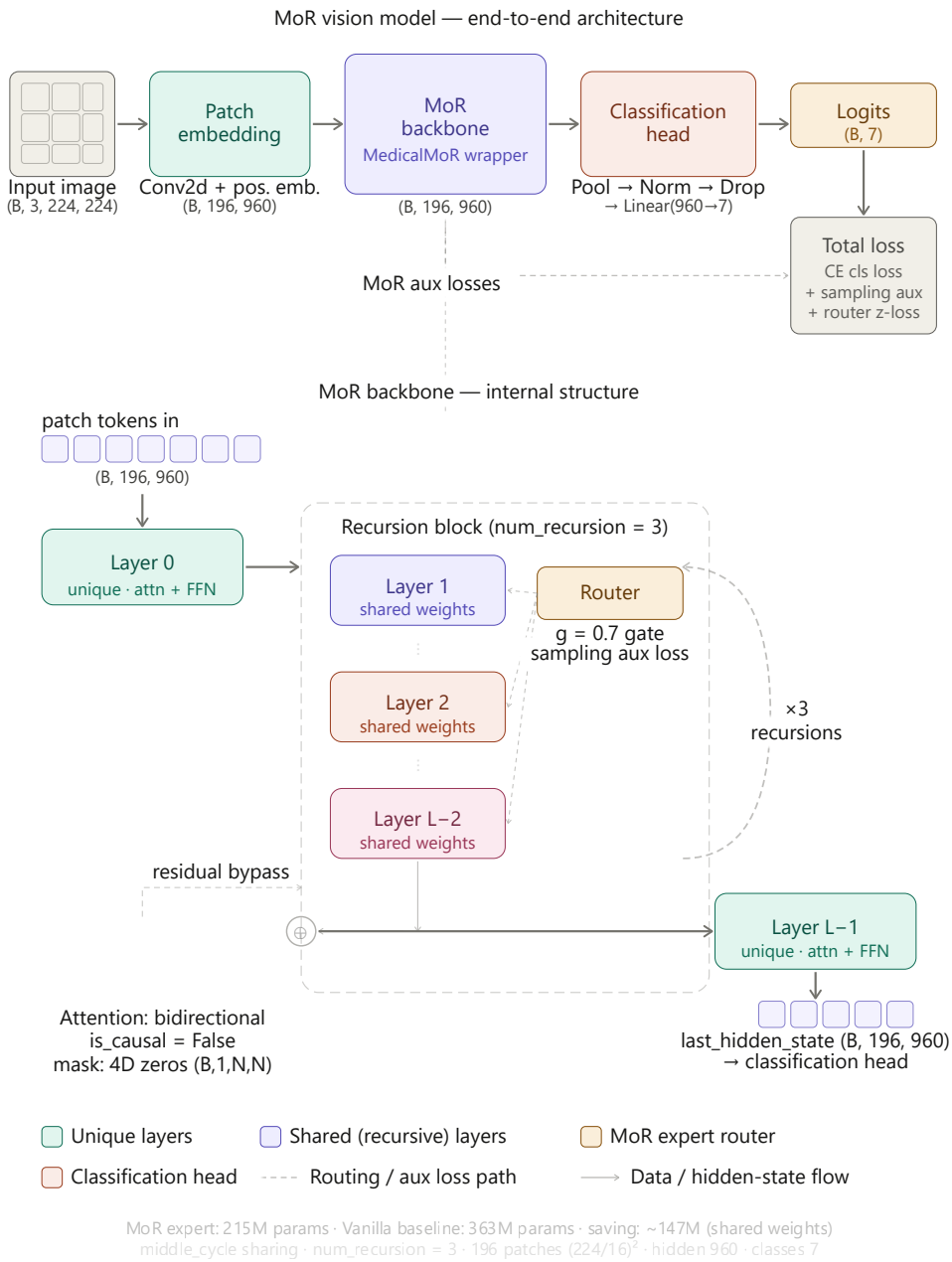


Figure 6.3: End-to-end vision architecture of the Mixture-of-Recursions (MoR) model, illustrating the pipeline from ViT-style image patch embedding through a recursively shared MoR transformer backbone with expert-choice routing, to global pooled feature aggregation and final seven-class HAM10000 classification output

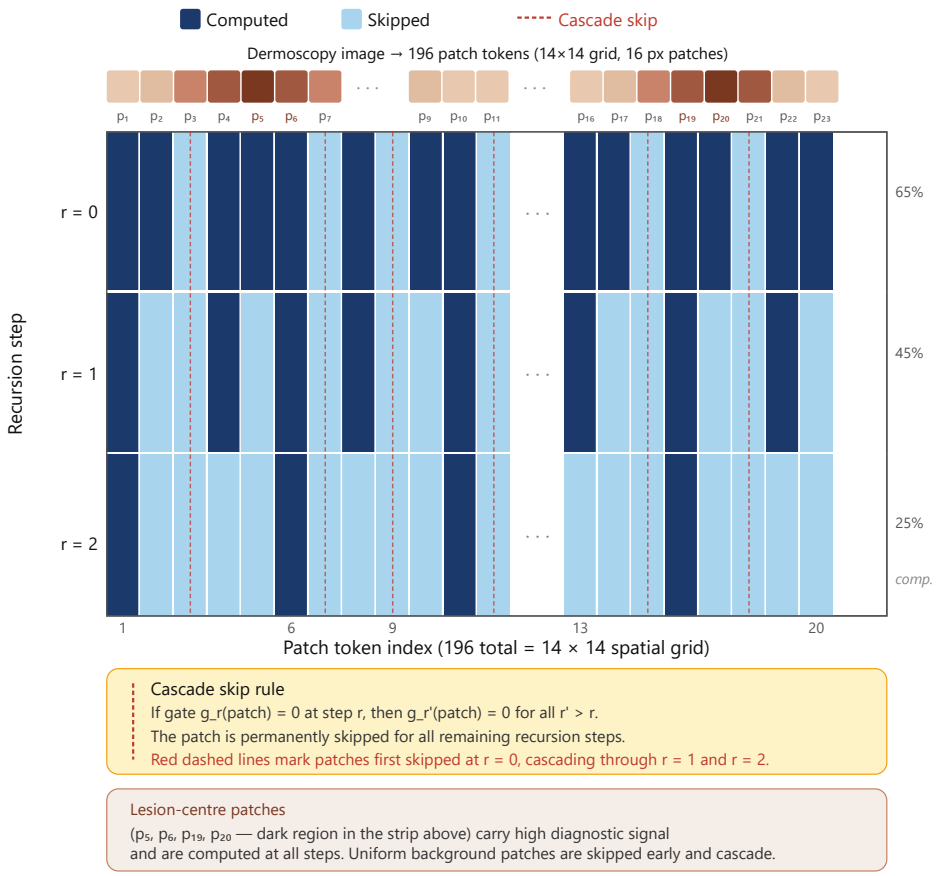


Figure 6.4: Adaptive router behaviour in the vision MoR backbone where informative image patch tokens receive deeper recursive processing while less salient patches are skipped.

6.4 ARCHITECTURAL RATIONALE AND TRADE-OFFS

The architectural choices in this dissertation are guided by a balance between controllability and expressiveness. A fully unconstrained adaptive model may produce richer behavior but can be difficult to diagnose. A heavily constrained model may be easy to analyze but too rigid to be useful. The selected MoR variants attempt to stay in the middle. Recursive depth is dynamic, but configuration surfaces such as recursion count, routing type, and gate controls remain explicit and tunable.

In the translation extension, the hybrid vanilla-encoder plus MoR-decoder design is

particularly pragmatic. It keeps source encoding stable and expressive while allowing adaptive recursion where uncertainty is most generation-sensitive, namely in decoder progression. This isolates adaptation to the part of the model where sequence formation errors are most visible and measurable through BLEU.

In vision, patch-based routing is a natural fit for interpretation because patches map back to image regions. This mapping does not automatically yield faithful explanation, but it provides an inspectable substrate. If depth increases consistently around ambiguous lesion boundaries or texture transitions, one can formulate concrete and testable interpretability claims tied to clinical reasoning patterns.

6.5 CAPACITY AND SHARING CONSIDERATIONS

Weight sharing and recursion depth are not merely efficiency choices; they influence representation dynamics. Sharing can regularize behavior and reduce parameter count, but it can also constrain specialization. Recursive gating can recover flexibility by modulating update strength across steps. The net behavior therefore emerges from interaction among shared transformations, gate calibration, and loss weights. This is why architecture must be interpreted jointly with optimization setup.

CHAPTER 7

LOSS FUNCTION

Loss design is central in adaptive routing because it determines whether gates remain informative or collapse to trivial behavior.

Depending on experimental configuration, the routing mechanism in this study operates in two related modes. In the machine translation extension, routing is implemented through soft continuous gating, where recursive contribution strength is modulated by learned scalar gate values. In contrast, major medical vision experiments employ sparse expert-choice routing based on capacity thresholds, where only a subset of patch tokens continues to deeper recursion steps. Both formulations share the same recursive backbone and routing objective, but differ in how computational depth is discretely or continuously allocated during training and inference.

7.1 CORE TASK LOSSES

7.1.1 Language Modeling Branch (Autoregressive)

For causal language modeling, the base objective is next-token prediction over non-masked positions:

$$\mathcal{L}_{\mathrm{LM}} = -\frac{1}{N_{\mathrm{tok}}} \sum_{t \in \mathcal{T}_{\mathrm{valid}}} \log p_\theta(x_t \mid x_{<t}),$$

where $N_{\mathrm{tok}} = |\mathcal{T}_{\mathrm{valid}}|$, and labels with ignore index are excluded.

In MoR pretraining, this base loss is augmented depending on router type (Section 7.2).

7.1.2 Machine Translation Branch (Seq2Seq with Label Smoothing)

For encoder–decoder translation, the sequence loss uses cross-entropy with label smoothing ε:

$$q_t(k) = \begin{cases} 1-\varepsilon, & k = y_t \\ \dfrac{\varepsilon}{|\mathcal{V}|-1}, & k \neq y_t \end{cases}, \qquad \mathcal{L}_{\text{MT,task}} = -\frac{1}{N_{\text{tok}}} \sum_{t \in \mathcal{T}_{\text{valid}}} \sum_{k \in \mathcal{V}} q_t(k) \log p_\theta(k \mid x, y_{<t}).$$

Representative configurations use $\varepsilon = 0.1$.

7.1.3 Vision Branch (Classification)

For medical image classification, the implemented task loss is:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{B} \sum_{b=1}^{B} \log p_\theta(y_b \mid I_b).$$

A weighted variant for imbalance handling is supported by project utilities:

$$\mathcal{L}_{\text{cls,w}} = -\frac{1}{B} \sum_{b=1}^{B} w_{y_b} \log p_\theta(y_b \mid I_b),$$

with class weights computed inversely to class frequency and normalized.

7.2 ROUTING-RELATED AUXILIARY TERMS

7.2.1 MT Extension: Decoder Gate Regularization

For translation with soft gates $g_{b,t}^{(r)} \in (0,1)$ at recursion step r, the total loss is:

$$\mathcal{L}_{\text{MT}} = \mathcal{L}_{\text{MT,task}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{ent}} \mathcal{L}_{\text{ent}} + \lambda_{\text{var}} \mathcal{L}_{\text{var}} + \lambda_{\text{tgt}} \mathcal{L}_{\text{tgt}}.$$

Balancing loss

$$\mu_r = \mathbb{E}_{b,t}\left[g_{b,t}^{(r)}\right], \qquad \mathcal{L}_{\text{bal}} = \text{Var}_r(\mu_r).$$

Entropy regularizer (binary gate entropy)

$$\mathcal{L}_{\text{ent}} = \frac{1}{R}\sum_{r=1}^{R}\mathbb{E}_{b,t}\left[-g_{b,t}^{(r)}\log g_{b,t}^{(r)} - \left(1 - g_{b,t}^{(r)}\right)\log\left(1 - g_{b,t}^{(r)}\right)\right].$$

Variance regularizer (token diversity)

$$\mathcal{L}_{\text{var}} = -\frac{1}{R}\sum_{r=1}^{R}\mathbb{E}_{b}\left[\text{Var}_{t}\left(g_{b,t}^{(r)}\right)\right].$$

Target-gate loss

$$\mathcal{L}_{\text{tgt}} = \frac{1}{R}\sum_{r=1}^{R}\left(\mathbb{E}_{b,t}\left[g_{b,t}^{(r),\text{preclamp}}\right] - \tau_r\right)^2,$$

where τ_r is a per-recursion target (e.g., [0.7, 0.5, 0.3]).

The target-gate regularization term is introduced to control recursion utilization across decoding stages. By encouraging the expected gate activation at each recursion step to match a predefined schedule (e.g., decreasing targets such as [0.7, 0.5, 0.3]), the model is guided toward a funnel-shaped compute allocation pattern. This helps prevent depth collapse in which all tokens receive either uniformly shallow or uniformly deep processing, and stabilizes training by maintaining meaningful diversity in recursive refinement. Consequently, the target schedule acts as a soft structural prior that promotes balanced adaptive computation without enforcing hard routing constraints.

Representative MT coefficients in project configs:

$$\lambda_{\text{bal}} = 0.01, \quad \lambda_{\text{ent}} = 0.0, \quad \lambda_{\text{var}} = 0.01, \quad \lambda_{\text{tgt}} \in \{0.05, 0.1\}.$$

7.2.2 LM MoR Auxiliary Terms (Project-Used Variants)

In language-model MoR training, the trainer composes the loss by router mode.

Expert router, `aux_loss` mode

$$\mathcal{L}_{\text{LM,total}} = \mathcal{L}_{\text{LM}} + \lambda_{\text{sam}}\,\mathcal{L}_{\text{sam}} + \lambda_z\,\mathcal{L}_z.$$

Here $\mathcal{L}_{\text{sam}}$ is BCE between router selection scores and selected-token targets, normalized by batch and sequence length.

Expert router, `aux_router` mode Main model optimization uses $\mathcal{L}_{\text{LM}} + \lambda_z \mathcal{L}_z$, while $\mathcal{L}_{\text{sam}}$ is optimized with a dedicated router update path.

Token router mode

$$\mathcal{L}_{\text{LM,total}} = \mathcal{L}_{\text{LM}} + \lambda_{\text{tok}}\,\mathcal{L}_{\text{tok_bal}} + \lambda_z\,\mathcal{L}_z,$$

with

$$\mathcal{L}_{\text{tok_bal}} = \sum_{i=1}^{R} P_i\, f_i, \quad P_i = \mathbb{E}_{b,t}[p_i(b,t)]\,, \quad f_i = R\,\rho_i, \quad \rho_i = \frac{1}{BT}\sum_{b,t} \mathbf{1}\left[\operatorname*{argmax}_j\; p_j(b,t) = i\right],$$

up to the implementation's batch-normalisation factor.

Router z-loss (used across variants when enabled)

$$\mathcal{L}_z = \mathbb{E}_{b,t}\left[\left(\log \sum_j e^{z_{b,t,j}}\right)^2\right]$$

(with implementation normalisation by effective items per batch).

Representative LM coefficients from project configs:

$$\lambda_{\text{sam}} = 10^{-3}, \quad \lambda_{\text{tok}} = 10^{-1}, \quad \lambda_z \in \{10^{-5},\, 10^{-3}\}.$$

7.2.3 Vision MoR Auxiliary Terms

Vision training combines classification with optional MoR penalties:

$$\mathcal{L}_{\text{vision,total}} = \mathcal{L}_{\text{cls}} + \lambda_{\text{sam}}\,\mathcal{L}_{\text{sam}} + \lambda_z\,\mathcal{L}_z.$$

In representative vision configs $\lambda_{\text{sam}} = 10^{-3}$, while z-loss is available but often disabled.

7.3 GATE CONSTRAINTS AND STABILIZERS

Router and gate constraints directly influence optimisation and interpretability:

$$g_{b,t}^{(r)} = \sigma\left(\frac{z_{b,t}^{(r)}}{T}\right), \qquad \tilde{g}_{b,t}^{(r)} = \text{clip}\left(g_{b,t}^{(r)},\ g_{\min},\ g_{\max}\right).$$

Representative settings include:

$$T = 1.0, \qquad g_{\min} = 0.05, \qquad g_{\max} = 0.95, \qquad \boldsymbol{\tau} = [0.7,\ 0.5,\ 0.3].$$

For cross-attention-aware MT routing, the gate logit input includes normalised cross-attention entropy:

$$\hat{H}_{b,t} = \frac{-\sum_s a_{b,t,s} \log a_{b,t,s}}{\log S},$$

which is concatenated with the decoder hidden state before the router MLP projection.

7.4 WHY LOSS DESIGN IS CENTRAL TO EXPLANATION

In adaptive routing systems, the loss function is part of the explanation mechanism itself. If auxiliary terms are too weak, routing may collapse to near-constant behaviour; if too strong, routing may become over-regularised and weakly coupled to task evidence.

For translation, $\mathcal{L}_{\text{bal}}$, $\mathcal{L}_{\text{var}}$, and $\mathcal{L}_{\text{tgt}}$ are anti-collapse controls that preserve depth diversity, while $\mathcal{L}_{\text{ent}}$ can sharpen decisions but may also bias gate distributions. For language-model MoR, sampling/balancing terms and z-loss similarly shape whether router activity is

informative or merely constrained. For vision, class imbalance handling and routing regularisation jointly affect whether depth maps track clinically meaningful uncertainty rather than dataset-frequency artefacts.

Therefore, routing claims are interpreted jointly with objective calibration:

$$\text{Interpretability quality} \propto \text{architecture} \times \text{loss calibration} \times \text{stability constraints}.$$

This motivates empirical validation of routing behaviour instead of inferring interpretability from design intent alone.

CHAPTER 8

EXPERIMENTAL SETUP

The setup is designed to compare explanatory behavior and transfer feasibility across three domains.

8.1 LANGUAGE MODELING EXPLANATORY EXPERIMENT

The language explanatory experiment uses recursion depth 3 with 50 sampled sequences and 102,400 token observations. Outputs include depth traces, entropy-related metrics, token identities, and router statistics.

8.2 MACHINE TRANSLATION EXPERIMENTS

Representative translation runs include an IWSLT17 de-en hybrid MoR 10k pilot, an IWSLT17 tiny overfit anti-collapse run, and an IITB en-hi hybrid vanilla-encoder plus MoR-decoder run.

The translation pipeline uses BLEU-centric callback evaluation, best-checkpoint saving, and optional best-checkpoint reload before final BLEU.

8.3 MEDICAL VISION EXPERIMENTS

Vision experiments target HAM10000 with MoR sparse variants and diagnostic routing analyses. Validation diagnostics include patch-depth distributions, and checkpoints track best validation behavior in sparse variants.

8.4 REPRODUCIBILITY AND TRACEABILITY PRACTICES

A key practical objective in this setup is traceability. Experimental conclusions should be reconstructable from stored artifacts without ambiguous interpretation. For this reason, the study emphasizes checkpoint-linked metrics, explicit evaluation intervals, and separation between periodic and final evaluation outputs in translation. In the vision branch, split counts and patch-depth diagnostics are explicitly tied to the analyzed validation path.

Another setup choice is to preserve weak outcomes rather than filtering only successful runs. This is methodologically important because instability patterns often reveal mechanism limitations that high-performing isolated runs do not capture. The translation branch is a clear example where variance itself is informative and helps prioritize stability-focused interventions.

CHAPTER 9

DATASETS

9.1 LANGUAGE DATA CONTEXT

The broader project includes FineWeb-Edu related language-model pretraining and evaluation workflows, while the explanatory module uses extracted token-level records from analyzed model behavior.

9.2 MACHINE TRANSLATION DATABASES

Major translation datasets used in this study context include IWSLT2017 de-en and IITB English-Hindi (`cfilt/iitb-english-hindi`). These provide contrast between smaller pilot-style and larger-scale translation regimes.

9.3 MEDICAL VISION DATABASE

HAM10000 is used for skin lesion classification with seven classes: `akiec`, `bcc`, `bkl`, `df`, `mel`, `nv`, and `vasc`. In analyzed splits, reported train and validation counts include 8,142 and 1,435 respectively in one pipeline path.

9.4 DATABASE CHARACTERISTICS AND THEIR INTERPRETIVE IMPACT

Table 9.1 summarises the key properties of the three datasets used across this study. Dataset characteristics directly influence routing interpretation. Language corpora with heavy frequency skew can naturally push routing toward lexical-prior behavior unless contextual signals are strong enough to override priors. Translation datasets differ in size, domain consistency, and sentence complexity, all of which affect gate stability and sequence-level outcomes. In medical vision, class imbalance and visual similarity between lesion types make uncertainty spatially structured rather than uniformly

distributed.

Table 9.1: Summary of datasets used across the three experimental domains.

Dataset	**Task**	**Size (train / val)**	**Imbalance**	**Metric**
FineWeb-Edu	Language modelling	102,400 observed tokens	Frequency skew	Entropy + depth
IWSLT2017 de-en	Machine translation	206,112 / 888	Moderate	BLEU
IITB en-hi	Machine translation	~1,660,000 / 520	Moderate	BLEU
HAM10000	Skin lesion classification	8,142 / 1,435	Severe	Accuracy

These characteristics are not noise around the model; they are part of the explanatory context. A robust interpretability argument must therefore account for database geometry, not treat it as a background detail.

CHAPTER 10

EXPERIMENTAL SETTINGS

10.1 SYSTEM AND HARDWARE CONFIGURATION

All experiments were conducted on a workstation equipped with an NVIDIA RTX A4500 GPU with approximately 20 GB of VRAM (20,470 MiB total capacity).

This hardware configuration enabled stable execution of recursive routing experiments, machine translation training cycles, and vision model convergence analysis without significant memory bottlenecks.

10.2 LANGUAGE-MODEL ROUTING SETTINGS

Language-model routing settings employ three recursion passes and a token-level audit performed at a large observation scale (102,400 tokens). The statistical analysis stack is centered on Spearman and Pearson correlation analysis, Mann–Whitney significance testing, and standardized effect-size interpretation to ensure robust behavioral diagnostics.

10.3 TRANSLATION SETTINGS

For the German–English machine translation experiments, the representative hybrid pilot configuration uses a hidden size of 576, encoder depth of 29 layers, decoder depth of 29 unrolled steps, and a recursion count of three. A cross-attention-aware routing mechanism is applied, combined with BLEU-based callback evaluation and best-checkpoint selection.

In the reported setup, the translation model was trained for approximately 10,000 training steps. Training and evaluation procedures include fixed logging intervals, periodic BLEU evaluation checkpoints, and early-stopping criteria based on validation

BLEU trends. Final evaluation runs may employ beam sizes different from those used during intermediate validation in order to assess terminal translation quality under inference-optimized conditions.

10.4 VISION SETTINGS

The vision experiments follow the seven-class HAM10000 dermatological classification setup with stratified dataset splitting and imbalance-aware handling strategies. Sparse expert-style Mixture-of-Recursions (MoR) variants are analyzed alongside patch-depth diagnostic signals derived from validation analysis notebook outputs.

Training of the vision model was subject to an early-stopping mechanism, which terminated optimization after 54 epochs when validation improvements plateaued. This stopping behavior is treated as an informative convergence signal rather than a purely procedural training constraint.

10.5 SETTING CHOICES AND THEIR CONSEQUENCES

The selected experimental configuration prioritizes interpretability, behavioral stability analysis, and transfer diagnostics over exhaustive hyperparameter optimization.

In language-model routing experiments, the large token observation scale and fixed recursion depth support reproducible profiling of routing dynamics. In translation experiments, the controlled training-step budget and structured checkpoint policy emphasize trajectory understanding rather than only final BLEU maximization. In vision experiments, early-stopping signals and patch-depth extraction are treated as primary analytical outputs alongside validation accuracy.

This configuration philosophy aligns with dissertation-stage objectives focused on mechanistic understanding and explanatory clarity. The resulting insights provide a principled basis for subsequent optimization phases, particularly in areas related to

calibration stability, routing efficiency, and cross-domain transfer robustness.

CHAPTER 11

RESULTS AND ANALYSIS

11.1 LANGUAGE-MODEL EXPLANATORY RESULTS

11.1.1 Depth Allocation Structure

Table 11.1: Depth allocation counts under MoR capacity constraints.

Depth	Count	Share (%)
Depth 1	34,150	33.35
Depth 2	34,150	33.35
Depth 3	34,100	33.30
Total	102,400	100.00

The routing distribution is near-uniform across depths, matching the intended cascade capacities and indicating stable load allocation rather than collapse to shallow routing.

11.1.2 Correlation Outcomes

Table 11.2: Correlation of recursion depth with uncertainty and token-level factors.

Metric	Spearman ρ	Spearman p	Pearson r	Pearson p
Pre-recursion entropy	-0.1389	0.00e+00	-0.0898	2.13e−182
Final entropy	+0.2306	0.00e+00	+0.2158	0.00e+00
Δ entropy ($H_{\text{pre}} - H_{\text{final}}$)	-0.2466	0.00e+00	-0.2424	0.00e+00
Log token frequency	-0.1907	0.00e+00	-0.1861	0.00e+00
Position	+0.0015	6.32e−01	+0.0015	6.32e−01

Interpretation.

1. A strong monotonic uncertainty→depth claim is not supported by pre-recursion entropy alone ($\rho = -0.1389$).

2. Frequency has a clearer role than initial entropy in routing depth assignment ($\rho = -0.1907$).

3. Position is effectively irrelevant ($\rho = 0.0015,\ p = 0.632$), indicating non-positional routing behavior.

4. The positive correlation with final entropy ($\rho = 0.2306$) suggests that tokens routed deeper remain globally difficult even after additional compute.

Table 11.3: Mean entropy profile by assigned depth.

Depth	Mean pre-recursion entropy	Mean final entropy	Mean Δ entropy
Depth 1	6.676	1.846	4.830
Depth 2	6.819	3.378	3.442
Depth 3	6.587	2.774	3.812

These means indicate that depth groups are not simply ordered by pre-recursion entropy; instead, routing appears to combine lexical priors and contextual signals.

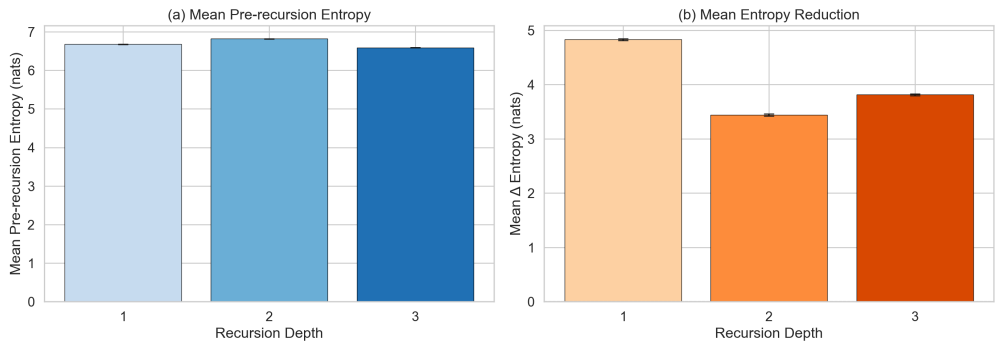


Figure 11.1: Relationship between assigned recursion depth and token-level entropy measures. The plot compares pre-recursion entropy, final entropy, and entropy reduction across depth groups. Contrary to the common assumption that higher uncertainty directly leads to deeper recursion, the observed pattern shows that depth is not monotonically aligned with pre-recursion entropy. Instead, deeper recursion is associated with tokens that remain difficult even after refinement, indicating that routing decisions reflect a combination of lexical priors and contextual signals rather than a single uncertainty metric. .

11.1.3 Category and Context Signals

Table 11.4: Mean depth by token category.

Category	Mean depth	Std. depth	Count	Mean pre-recursion entropy
Alphabetic	2.0767	0.8201	80,270	6.6549
Digit	2.0296	0.8015	6,796	6.6983
Mixed	1.9254	0.7476	536	6.9598
Punctuation	1.6109	0.6472	12,972	6.9385
Whitespace	1.2892	0.6584	1,739	6.5540
Special	1.0000	0.0000	87	7.1008

For alphabetic tokens, subword continuations are routed shallower than word-initial pieces:

$$\text{initial_alphabetic} = 2.0957, \quad \text{subword_alphabetic} = 1.9882.$$

This is consistent with predictability intuition and supports the claim that continuation fragments are generally easier.

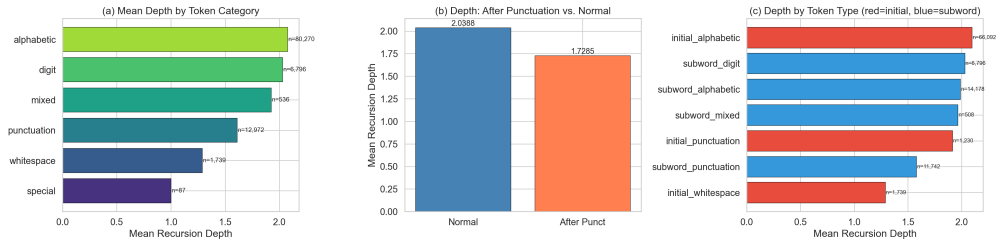


Figure 11.2: Average recursion depth across different token categories, including alphabetic, numeric, punctuation, whitespace, and special tokens. The results show that linguistically meaningful tokens (e.g., alphabetic tokens) receive deeper recursive processing on average, while structurally predictable elements such as whitespace and special tokens are processed at shallower depths. This supports the hypothesis that routing captures token-level complexity beyond simple frequency effects, aligning partially with predictability-based interpretations of adaptive computation.

11.1.4 Router Separation Strength

A total of 54,610 router-score records were collected across routing steps.

Table 11.5: Selected vs. unselected router-score separation at deeper routing steps.

Step	Selected mean	Unselected mean	Gap
Step 2	0.9957	0.0222	0.9735
Step 3	0.9932	0.0149	0.9783

The near-binary score gaps indicate highly decisive token selection once routing proceeds beyond the first mandatory pass.

11.1.5 Counterfactual and Consistency

Table 11.6: Frequency-only counterfactual agreement with actual routing.

Metric	Value
Exact match rate	0.3877 (38.8%)
Cohen's κ	0.0815
Spearman(actual depth, counterfactual depth)	0.1792

Table 11.7: Counterfactual confusion matrix (rows: actual depth, columns: frequency-only depth).

Actual \ Counterfactual	Depth 1	Depth 2	Depth 3
Depth 1	11,391	10,426	12,333
Depth 2	18,650	11,019	4,481
Depth 3	4,109	12,705	17,286

Table 11.8: Per-depth agreement with frequency-only heuristic.

Depth	Agreement (%)	Support
Depth 1	33.3	34,150
Depth 2	32.3	34,150
Depth 3	50.8	34,100

Table 11.9: Within-token depth consistency (modal-depth fraction).

Statistic	Value
Mean modal consistency	0.7535
Median modal consistency	0.7619

Interpretation.

1. Frequency explains part of routing (38.8% exact agreement), but low $\kappa = 0.0815$ shows weak beyond-chance agreement overall.

2. The heuristic matches depth 3 better than depths 1–2, indicating that rarity contributes most to identifying high-compute tokens.

3. Mean modal consistency of 0.7535 indicates non-trivial stability for a token's preferred depth, while leaving substantial room for context-dependent reassignment.

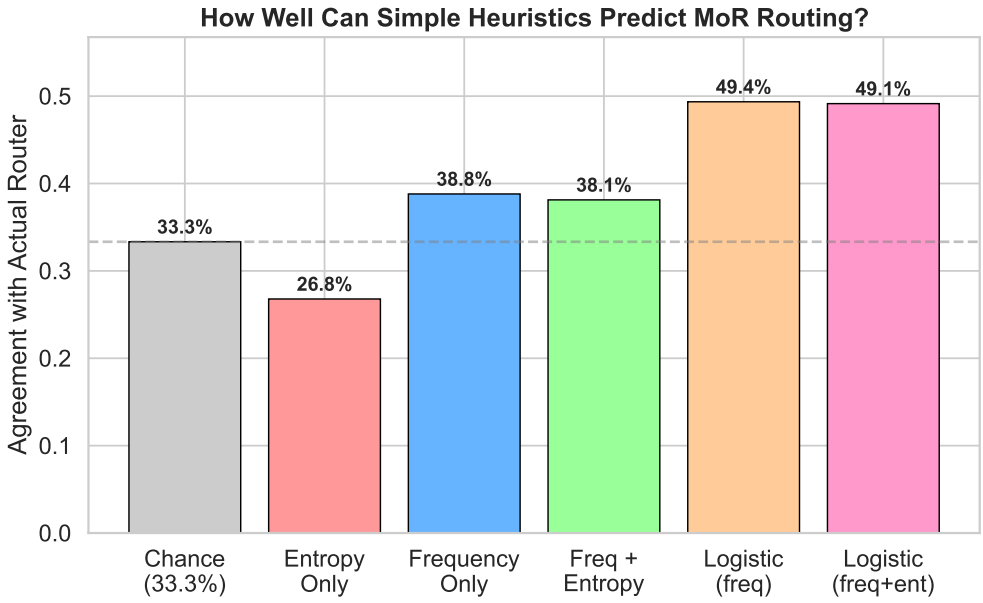


Figure 11.3: Comparison of actual routing decisions with predictions from simple heuristic models, including entropy-only, frequency-only, and logistic combinations of features. The figure shows that while frequency-based heuristics provide some predictive power, they fail to fully explain routing behavior. The gap between heuristic performance and actual routing indicates that the model leverages additional contextual and learned signals, supporting the claim that MoR routing is non-trivial and not reducible to simple token-level statistics.

11.2 MACHINE TRANSLATION RESULTS

11.2.1 Machine Translation Results and Interpretation

The representative translation performance obtained in the hybrid encoder–decoder MoR setting corresponds to a BLEU score of **29.44** on the IWSLT17 *German–to–English* (de→en) task. This result was achieved using the hybrid configuration consisting of a

vanilla Transformer encoder and a recursively refined MoR decoder with middle-cycle parameter sharing.

An additional set of preliminary experiments was conducted on the *English–Hindi* translation setting using the IITB corpus. Although the architectural pipeline was successfully extended to this linguistically distinct and morphologically richer task, the resulting translation quality remained significantly below acceptable performance levels under the explored configurations. Consequently, further optimization in this direction was deferred in order to focus on the more stable German–English evaluation setting.

From an interpretational perspective, the observed BLEU score on the IWSLT17 benchmark indicates that the proposed hybrid MoR extension is functionally capable of supporting sequence-to-sequence generation with adaptive recursive computation in the decoder. The result demonstrates that parameter sharing combined with token-level adaptive depth does not prevent the model from learning meaningful alignment and fluency patterns in a medium-scale translation scenario.

At the same time, the sensitivity observed during multilingual extension experiments suggests that routing dynamics, recursion depth allocation, and optimization stability play a critical role in determining downstream translation quality. This highlights that while the architectural framework is operationally complete, achieving consistently strong performance requires careful tuning of training schedules, data scale, and recursion configuration, particularly in low-resource or high-morphology language pairs.

11.3 QUANTITATIVE MEDICAL VISION RESULTS

11.3.1 Baseline and Pretraining Comparisons

To evaluate the performance and efficiency of the Mixture of Recursions (MoR) architecture in a medical vision context, the model was compared against a standard Vision Transformer (ViT) baseline. When initialized with pretrained ViT-Huge IN21k

weights, the Vanilla baseline achieves a validation accuracy of 0.8976 utilizing 631M parameters. In contrast, the pretrained MoR model, using the same initialization strategy and a recursion depth of 3, achieves an accuracy of 0.8209 with only 237M parameters.This demonstrates a significant efficiency advantage: MoR operates with 394M fewer parameters than the Vanilla model. While this parameter reduction incurs a 7.7 percentage point accuracy cost, MoR uniquely produces spatially interpretable patch-depth maps without any additional supervision—a diagnostic signal that the Vanilla architecture cannot provide. Additionally, utilizing pretrained weights provides an approximate +0.06 accuracy improvement over the best-performing from-scratch training configuration, highlighting the importance of initialization in adaptive routing stability.

11.3.2 Recursion Depth Ablation Study

An ablation study was conducted on the from-scratch MoR variants to analyze the impact of varying the maximum recursion depth (2, 3, and 5) on both parameter count and classification performance.

- A maximum recursion depth of 2 achieves a validation accuracy of 0.7721 with 252M parameters.
- A maximum recursion depth of 3 achieves a validation accuracy of 0.7617 with 215M parameters.
- A maximum recursion depth of 5 achieves a validation accuracy of 0.7443 with 185M parameters.

The results reveal a clear architectural trade-off. As the recursion count increases, the total parameter footprint decreases due to more extensive weight sharing across the recursive steps. However, in the from-scratch training regime, higher recursion depths correlate with a slight degradation in validation accuracy, indicating that deeper unrolled routing requires precise gate calibration to maintain optimization stability.

Model	Backbone Init	Rec.	Val Acc.	Epoch Stopped	Params
MoR (scratch)	NLP backbone	3	0.7617	26	215M
MoR (scratch)	NLP backbone	2	**0.7721**	37	252M
MoR (scratch)	NLP backbone	5	0.7443	23	185M
MoR (pretrained)	ViT-Huge IN21k	3	**0.8209**	52	237M
Vanilla (pretrained)	ViT-Huge IN21k	—	**0.8976**	43	631M

11.3.3 Vision Explainability Interpretation

The vision routing maps are particularly promising for explanation. The observed pattern is that higher recursion often appears around lesion-relevant neighborhoods and transition zones, while lower recursion appears in more homogeneous areas where the model appears more confident.

A clinically meaningful interpretation is that MoR may be allocating extra compute to ambiguous boundaries, texture transitions, and context around diseased structures, while avoiding unnecessary depth in visually simpler regions.

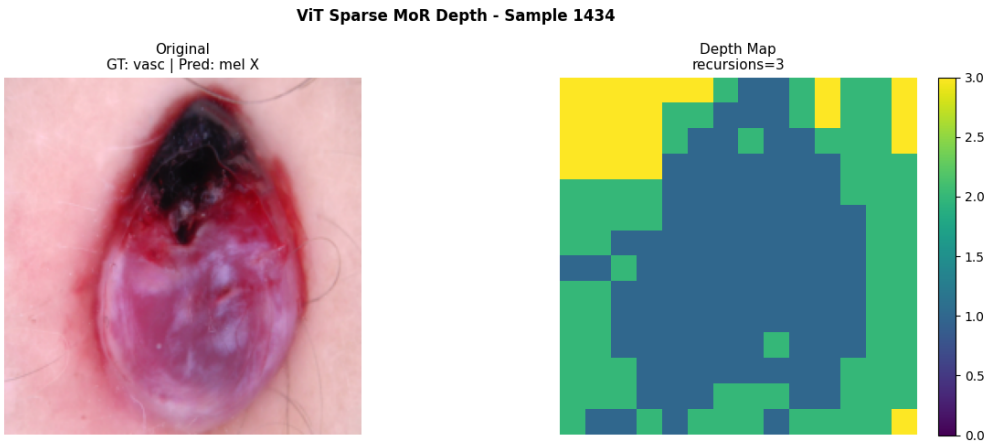


Figure 11.4: Patch-level recursion depth map for Sample 1434. The visualization illustrates the model's adaptive compute allocation: the concentrated, distinct core of the lesion receives lower recursion depth (dark blue), indicating high model confidence and visual homogeneity. In contrast, the model allocates higher recursive steps (green and yellow) to the surrounding boundaries and transition zones, signaling localized uncertainty and a demand for feature refinement where the disease merges with healthy tissue.

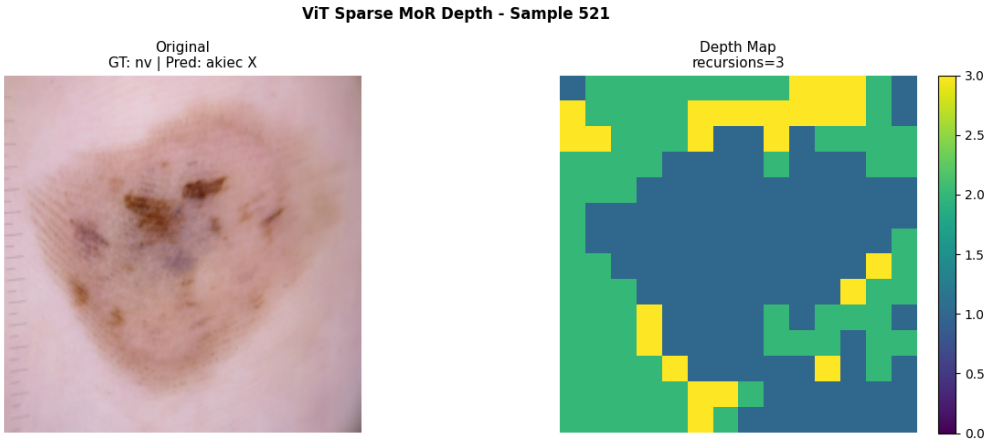


Figure 11.5: Patch-level recursion depth map for Sample 521. This sample further demonstrates the refinement-demand behavior of the MoR routing mechanism. Computational effort is minimized within the primary disease concentration due to its predictable structure, while the ambiguous texture transitions and peripheral areas around the lesion trigger deeper recursion to resolve structural uncertainty.

This behavior is valuable because lesion diagnosis is often not determined by one isolated pixel region. Border irregularity, surrounding skin context, and local contrast transitions can all carry diagnostic signal. Depth concentration around these regions suggests an explanation mechanism aligned with diagnostic reasoning.

A second important point is that some lesion-core areas may receive lower depth than nearby boundary and context patches. This is not necessarily contradictory. If the lesion core is visually uniform or already confidently recognized, the model may spend more recursive effort on difficult border and context patches where class-discriminative uncertainty is higher. The depth map can therefore be read as indicating where the model still needs refinement, not simply where disease exists. This is a stronger interpretability story.

Property	**Vanilla**	**MoR**
Spatial effort signal	✗	✓
Per-patch depth map	✗	✓
Boundary uncertainty	✗	✓
No extra supervision	✓	✓
No architectural change	✓	✓

Clinical potential: Boundary delineation is the hardest part of dermoscopic diagnosis. MoR surfaces this *for free* as a byproduct of routing.

Multi-Depth Routing Dynamics

Analyzing the patch-level depth maps across different maximum recursion bounds further validates the model's refinement-demand behavior. When observing the same dermoscopic images routed through configurations with maximum depths of 2, 3, and 5, the routing maps shift dynamically to utilize the available computational budget. Deeper recursion configurations resolve finer spatial structures within the image. As the allowed recursion increases to 5, the model achieves sharper core isolation, concentrating the lowest-depth computations strictly on the most diagnostically certain and visually uniform areas of the lesion. Consequently, the highest recursion depths are pushed outward to the most ambiguous boundaries and peripheral textures, confirming that MoR actively scales its local spatial refinement based on structural uncertainty.

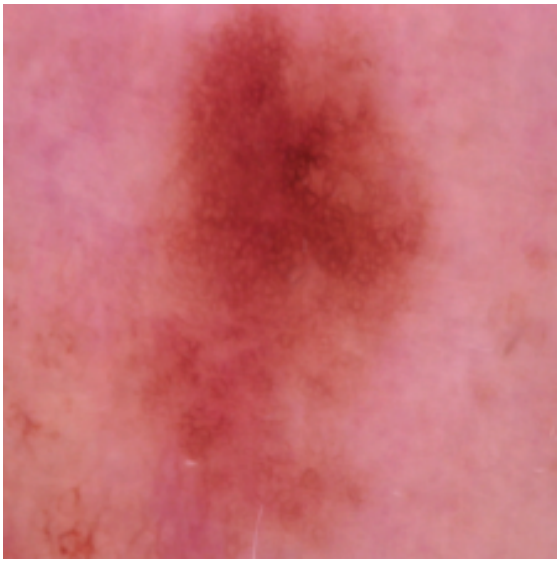
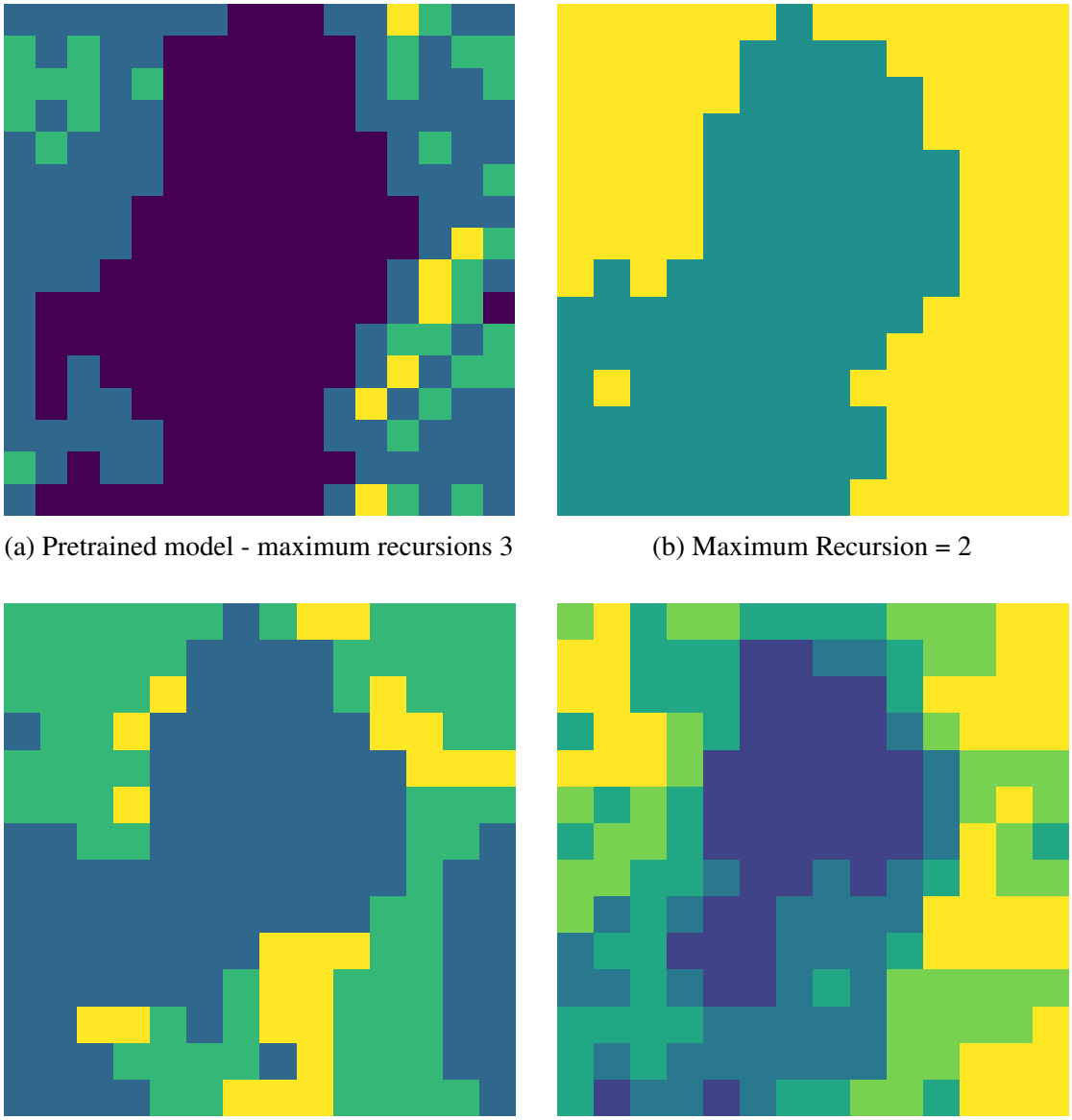


Figure 11.6: Original image



(a) Pretrained model - maximum recursions 3

(b) Maximum Recursion = 2

(c) Maximum Recursion = 3

(d) Maximum Recursion = 5

Figure 11.7: Ablation of maximum recursion depth on patch-level routing dynamics. As the allowed recursion budget increases from 2 to 5, the model resolves finer spatial structures, achieving sharper core isolation (dark blue) and concentrating the highest recursion depths on the most ambiguous boundary and transition textures.

11.4 INTEGRATED CROSS-DOMAIN READING OF THE RESULTS

Taken together, the three result streams suggest that MoR has reached a meaningful intermediate maturity point. Routing is clearly active and structured, transfer is feasible,

and explanation is possible in task-specific ways. At the same time, the evidence argues against simplistic universal claims. In language, depth cannot be treated as a direct monotonic proxy for pre-recursion uncertainty in the analyzed run. In translation, performance quality is achievable but not uniformly stable. In vision, explainability behavior is promising but still needs stronger class-wise and calibration-backed validation.

The integrated reading is therefore balanced. Structural competence appears before full semantic confidence. This pattern is common in adaptive systems: the model learns to make confident allocation decisions first, and only later do those decisions align consistently with the exact explanatory proxy researchers initially expected.

11.5 WHY THE VISION SIGNAL IS ESPECIALLY IMPORTANT

Among the three branches, the vision signal is especially important for interpretability discourse because it is spatially inspectable. Token-level language routing is informative but abstract to non-specialists. Patch-level depth maps, in contrast, can be related to visible lesion structures and boundary complexity. This creates a direct bridge between model internals and human reasoning.

The observation that higher depth appears around diseased neighborhoods while some lesion-core areas receive lower depth can be interpreted in a principled way. In many dermoscopic images, lesion centers may be visually homogeneous once identified, whereas boundaries and nearby context contain irregularity, transition texture, or color shifts that drive class ambiguity. A refinement-seeking router would naturally allocate additional recursion to these uncertain transition regions. Under this view, low depth in some lesion-core areas is not failure; it can indicate confident recognition with compute saved for harder surrounding structure.

This interpretation aligns with the concept of adaptive compute as localized uncertainty resolution. It also suggests a concrete evaluation extension: comparing patch-depth

concentration with boundary-centric uncertainty or disagreement maps. If depth consistently aligns with these maps, the explanatory claim becomes significantly stronger.

11.6 PRACTICAL INTERPRETATION OF TRANSLATION VARIANCE

The spread of BLEU outcomes should be interpreted diagnostically. A high-performing pilot alongside weak runs implies the architecture can reach good solutions but does not do so robustly across settings. This usually indicates sensitivity to schedule, regularization balance, and evaluation policy. The existing infrastructure, especially best-checkpoint handling and final BLEU procedures, is already sufficient to support a systematic stabilization program.

From an explanatory perspective, this matters because unstable optimization can mask routing semantics. If gates are not stably trained, interpretability analyses become run-fragile. Improving translation stability is therefore not only a performance objective but also an interpretability objective.

11.7 STRENGTH OF THE LANGUAGE EXPLANATORY MODULE

The language module remains the methodological core of the thesis. Its strength lies in refusing a single-axis interpretation. Instead of concluding from one correlation, it combines depth distributions, category behavior, router score separation, counterfactual mismatch, and consistency measures. The result is a nuanced but credible account: routing is meaningful and structured, yet its semantics are composite rather than one-dimensional.

CHAPTER 12

DISCUSSION

This study changes how MoR should be presented academically.

12.1 MAIN INTERPRETATION

MoR should be described as a constrained adaptive-allocation mechanism whose behavior emerges from the interaction of routing policy, capacity geometry, auxiliary regularization, and optimization dynamics.

12.2 CLAIM-LEVEL SUMMARY

Supported claims include structured depth allocation under constraints, rare-token association with deeper recursion, strong router separation at deeper steps, near-independence of depth from token position, and active non-trivial depth routing in vision. Partially supported claims include context-sensitive depth assignment for repeated identities and difficulty alignment that depends on proxy definition and setting. Not supported in the analyzed NLP run are the monotonic positive depth versus pre-recursion entropy claim and the claim that the first token after punctuation receives deeper recursion.

12.3 WHY THE VISION RESULT IS STRONG FOR EXPLANATION

The vision extension provides one of the clearest practical explanation pathways in the thesis. Depth is spatially visualizable per patch, routing naturally exposes where compute is concentrated, and concentration around lesion-adjacent and transition regions supports an interpretable narrative of refinement under uncertainty.

Crucially, this spatial allocation mechanism closely mirrors the diagnostic reasoning of a human dermatologist. When examining a lesion, a clinician quickly processes visually uniform areas—whether that is the unmistakable, homogenous core of a disease or the clear, healthy surrounding skin. Their intensive cognitive effort is instead concentrated on the ambiguous margins, irregular borders, and texture transitions where the diagnosis is ultimately determined. The MoR model exhibits the exact same behavior: it allocates the lowest computational depth to the most concentrated, unambiguous regions of the disease and homogeneous healthy skin, while reserving its highest recursion depths for the less concentrated, highly ambiguous transition zones.

Furthermore, as the maximum recursion limit increases, the model becomes increasingly expressive. Higher recursion budgets allow the architecture to resolve these spatial diagnostic boundaries with much sharper precision, mapping directly to clinical regions of interest.

However, this increased spatial expressiveness currently involves a trade-off, as validation accuracy slightly degrades at higher recursion depths when trained from scratch. This indicates that deeper unrolled routing introduces optimization instabilities. Yet, this presents a clear and highly promising path forward. If future architectural modifications such as improved auxiliary loss calibration, advanced gate stabilization, or dedicated medical pretraining can close this minor accuracy gap, this visually grounded, refinement-based MoR architecture has immense potential for practical, trustworthy deployment in clinical diagnostic settings. This is exactly the kind of evidence that can bridge model internals and human-understandable explanations, presenting a compelling explainability direction while still noting that clinical grade deployment requires strict class-wise and calibration analyses.

12.4 LIMITATIONS

Some conclusions are run-specific and require multi-seed confirmation. Statistical significance in large samples can overstate practical magnitude. Metadata parity is incomplete across all vision run families, and MT comparability depends on strict evaluation-policy standardization.

12.5 PRACTICAL NEXT STEPS

The immediate practical next steps are to standardize result schemas across tasks, enforce mandatory checkpoint metadata fields, add seed-averaged benchmark tables, export joint quantitative and routing-map reports for vision, and include per-class and calibration metrics for medical settings.

CHAPTER 13

CONCLUSION

This thesis presents MoR as a study of interpretable adaptive allocation, not merely as a model implementation exercise. The work first establishes an explanatory module for routing behavior, then extends MoR to machine translation and medical vision.

The language-model audit reveals a mature and nuanced result: routing is clearly structured and meaningful in several dimensions, but the strongest uncertainty-monotonic narrative is not supported in the analyzed run. This finding improves, rather than weakens, scientific understanding by clarifying what MoR depth appears to encode.

The machine translation extension demonstrates architectural and instrumentation feasibility with BLEU-aware checkpoint logic, while also revealing stability sensitivity that now has clear diagnostic pathways.

The medical vision extension is especially encouraging. Both quantitative outcomes and spatial depth behavior suggest that MoR can provide a practical explanatory signal: extra recursion where ambiguity is likely higher, reduced recursion where confidence appears higher. This aligns well with the idea of compute as explanation.

An important conceptual refinement follows from this study: recursion depth should not be interpreted as a binary disease detector. It is better understood as a refinement-allocation signal. In vision, this means depth may rise around uncertain lesion boundaries and difficult contextual transitions even when some lesion-core regions receive lower depth. In language, this means depth reflects a blend of priors and context, not only one uncertainty statistic.

This refined interpretation is scientifically stronger because it aligns with observed

behavior across tasks. It also provides a practical foundation for future explainable AI work: if compute allocation can be measured as refinement demand, then depth maps and routing trajectories become analyzable explanatory artifacts rather than post-hoc visualizations.

The next stage of this dissertation line is therefore clear. Maintain the study-first stance, expand seed-level robustness, standardize artifacts, and deepen calibration-aware analysis, especially in medical vision where explanatory usefulness can have high practical impact. With these steps, MoR can evolve from a promising adaptive mechanism to a mature and interpretable computation framework.

The overall conclusion is balanced and actionable. MoR is a strong candidate for adaptive, interpretable computation, but its interpretability must be tested claim-by-claim rather than assumed. Vision routing behavior, in particular, offers a promising bridge to explanation-centric AI when combined with rigorous clinical evaluation protocols.

BIBLIOGRAPHY

Bae, S., **Y. Kim**, **R. Bayat**, *et al.* (2025). Mixture-of-recursions: Learning dynamic recursive depths for adaptive token-level computation. *arXiv preprint arXiv:2507.10524*.

Dosovitskiy, A., **L. Beyer**, **A. Kolesnikov**, **D. Weissenborn**, **X. Zhai**, **T. Unterthiner**, **M. Dehghani**, **M. Minderer**, **G. Heigold**, **S. Gelly**, **J. Uszkoreit**, and **N. Houlsby** (2021). An image is worth 16x16 words: Transformers for image recognition at scale. *International Conference on Learning Representations (ICLR)*.

Graves, A. (2016). Adaptive computation time for recurrent neural networks. *arXiv preprint arXiv:1603.08983*.

Shazeer, N., **A. Mirhoseini**, **K. Maziarz**, *et al.* (2017). Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *International Conference on Learning Representations*.

Teerapittayanon, S., **B. McDanel**, and **H. T. Kung**, Branchynet: Fast inference via early exiting from deep neural networks. *In International Conference on Pattern Recognition*. 2016.

Tschandl, P., **C. Rosendahl**, and **H. Kittler** (2018). The ham10000 dataset: A large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, **5**, 180161.

Vaswani, A., **N. Shazeer**, **N. Parmar**, **J. Uszkoreit**, **L. Jones**, **A. N. Gomez**, **L. Kaiser**, and **I. Polosukhin** (2017). Attention is all you need. *Advances in Neural Information Processing Systems (NeurIPS)*.

Zhang, D., **J. Song**, **Z. Bi**, **X. Song**, **Y. Yuan**, **T. Wang**, **J. Yeong**, and **J. Hao** (2024). Mixture of experts in large language models. *arXiv preprint arXiv*.